MTH207M (2023-24, ODD SEMESTER)

PROBLEM SET 4

1. Consider the model $E(y_1) = \beta_1 + \beta_2$, $E(y_2) = \beta_1 - \beta_2$, $E(y_3) = \beta_1 + 2\beta_2$ with the usual assumptions. Obtain the BLUE of $2\beta_1 + \beta_2$ and find its variance. Also, find $\hat{\beta}$.

2. Consider the model $E(y_1) = 2\beta_1 + \beta_2$, $E(y_2) = \beta_1 - \beta_2$, $E(y_3) = \beta_1 + \alpha\beta_2$ with the usual assumptions. Determine α such that the BLUEs of β_1, β_2 are uncorrelated.

3. Consider the model $E(y_1) = \beta_1 + \beta_2$, $E(y_2) = 2\beta_1$, $E(y_3) = \beta_1 - \beta_2$ with the usual assumptions. Find the RSS.

4. Consider the one-way Anova model:

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij},\ i = 1, \ldots, k,\ j = 1, \ldots, n_i,$$

   where ϵ_{ij} are independent with the mean 0 and variance σ^2. What are the estimable functions? Is the grand mean $\bar{y}$ an unbiased estimator of μ?

5. Consider the model $E(y_1) = \beta_1 + 2\beta_2$, $E(y_2) = 2\beta_1$, $E(y_3) = \beta_1 + \beta_2$ with the usual assumptions. Find the RSS subject to the restriction $\beta_1 = \beta_2$.

6. Let $x_1, \ldots, x_n$ be real numbers with mean $\bar{x}$. Consider the linear model

$$y_i = \alpha + \beta(x_i - \bar{x})$$

   with the usual assumptions. Show that the BLUEs of α and β are uncorrelated.

7. Prove that the BLUE of an estimable function is unique. (It is to be shown that if $l^T\beta$ is an estimable function and if $c^T y$ and $d^T y$ are both BLUE of the function, then $c = d$.)